

Данное задание ~1
 вариант ~96
 ~1

$$A = 375$$

а). BCD

$$\underbrace{0000.0011}_3 \quad \underbrace{0111.0101}_{4 \quad 5}$$

б). ASCII

$$\underbrace{0011.0011}_3 \quad \underbrace{0011.0111}_7 \quad \underbrace{0011.0101}_5$$

~2

$$A = 375 \quad -A = -375$$

$$A_2 = 1.0111.0111$$

$$0000.0001.0111.0111$$

$$[-A]_p = 1.000.0001.0111.0111$$

$$[-A]_5 = 1.111.1110.1000.1000$$

+

$$[-A]_{\text{gen}} = \begin{array}{r} 1.111.1110.1000.1000 \\ \hline 1.111.1110.1000.1001 \end{array}$$

$\overbrace{11000011}^C \overbrace{10101001}^A \overbrace{11000001}^E \overbrace{10000000}^1 \overbrace{0000}^0$
 3 байта ХАРАКТЕРИСТИКА МАНИССА

$$P_V = X_V - 128 = 67 - 128 = -61$$

$$V = -(0,1001000111000001)_2 \cdot 2^7$$

$$V = -(1010100,111000001)_2 \approx -84,88$$

$\overbrace{0011}^3 \overbrace{1101}^2 \overbrace{0100}^4 \overbrace{1100}^C \overbrace{0000}^0 \overbrace{0000}^0 \overbrace{0000}^0 \overbrace{0000}^0$
 3 байта ХАРАКТЕРИСТИКА МАНИССА

$$P_W = X_W - 128 = 122 - 128 = -6$$

$$W = (0,110011)_2 \cdot 2^6$$

$$W = (0,000000110011)_2 \approx 0,0125$$

~8

$$T = ? \quad R = \text{C3ASC100}$$

$$Q = ? \quad S = 3D4C0000$$

~7
 Требуется найти - см. ~7

$$P_T = X_T - 127 = 135 - 127 = 8$$

$$T = -(\overbrace{10011000}^{\text{deleted}} - (1,010100111000001)_2 \cdot 2^8$$

$$T = -(101010011,1000001)_2 \approx -339,5$$

$$P_Q = X_Q - 127 = 122 - 127 = -5$$

$$Q = (1,10011)_2 \cdot 2^{-5}$$

$$Q = (0,0000110011)_2 \approx 0,498$$

$$X_A = P_A + 127 = 135 = (10000111)_2$$

$$\underbrace{0}_{\text{знак}} \underbrace{10000111}_{\text{смешанный порядок}} \underbrace{011101110000000000000000}_{\text{мантисса}}$$

$$B = (0,074)_{10} = (0,12F1A9FC)_{16} = (0,00010010111100011010100111111100)_2 =$$

$$= (1,00101111000110101001111111)_2 \cdot 2^{-4}$$

$$X_B = P_B + 127 = 123 = (01111011)_2$$

$$\underline{0,01111011,00101111000110101001111111}$$

$$Y = ? \quad R = C3A9C100 \quad \sim 6$$

$$Z = ? \quad S = 3D4C0000$$

$$R = \underbrace{1100}_{\text{знак}} \underbrace{0011}_{\text{характеристика}} \underbrace{101010011100000100000000}_{\text{мантисса}}$$

$$X_Y = 67 = \underbrace{64}_{\text{смеш. порядок}} + \underbrace{3}$$

$$P_Y = X_Y - 64 = 3$$

$$Y = -(C, A9C1)_{16} \cdot 16^3$$

$$Y = -(A9C, 1)_{16} = -2716,0625$$

$$S = \underbrace{0011110101001100}_{\text{знак}} \underbrace{0000000000000000}_{\text{мантисса}}$$

$$P_Z = X_Z - 64 = 61 - 64 = -3$$

$$Z = (0,4C)_{16} \cdot 16^3 = (C,004C)_{16} \approx 0,00007248$$

~ 7

$$V = ? \quad R = C3A9C100$$

$$W = ? \quad S = 3D4C0000$$

~3

$$A=375; B=0,074$$

$$A = (375)_{10} = (177)_{16} = (0,177)_{16} \cdot 16^3$$

$$X_A = P_A + 64 = (67)_{10} = (1000011)_2$$

$$\begin{array}{ccccccc} 0 & 100 & 0011 & 0001 & 0111 & 0111 & 0000 & 0000 & 0000 \\ + & \text{ХАРАКТЕРИСТИКА} & & & & & \text{МАНТИССА} & & \end{array}$$

$$B = (0,074)_{10} = (0,12F1AA)_{16} = (0,12F1AA) \cdot 16^0$$

$$X_B = P_B + 64 = 0 + 64 = 64 = (1000000)_{10}$$

$$0,100 \quad 0000,0001 \quad 0010 \quad 1111 \quad 0001 \quad 1010 \quad 1010$$

~4

$$A=375; B=0,074$$

$$A = (375)_{10} = (177)_{16} = (101110111)_2 = (0,101110111) \cdot 2^9$$

$$X_A = P_A + 128 = 137 = (10001001)_2$$

$$\begin{array}{ccccccc} 0,10001001,101110111000000000000000 \\ + & \text{ХАРАКТЕРИСТИКА} & & \text{МАНТИССА} & & & \end{array}$$

$$B = (0,074)_{10} = (0,12F1A9FB)_{16} = (0,00010010 \quad 1111 \quad 0001 \quad 1010 \quad 1001 \quad 1111 \quad 1100)_2 =$$

1 2 F 1 A 9 F C

$$= (0,10010111100011010100111111100)_2 \cdot 2^{-3}$$

$$X_B = P_B + 128 = 125 = (01111101)_2$$

$$\begin{array}{ccccccc} 0,01111101,010010111100011010100111111 \\ + & \text{ХАРАКТЕРИСТИКА} & & \text{МАНТИССА} & & & \end{array}$$

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$$A=375; B=0,074$$

$$A = (375)_{10} = (177)_{16} = (101110111)_2 = (1,01110111) \cdot 2^8$$